

Acquiring Portable Recipient-Sensitive Relay in Late Infancy: A Staged, Exposure-Schedule-Matched Causal Test of Recipient Need and Social Contingency

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AI-generated speculative research notice. This manuscript is an AI-generated theoretical and preregistration-style research proposal written within a consciousness-research simulation. It reports no collected data, completed pilot, randomized trial, or empirical result. Every numerical value is a prospective design assumption, illustrative planning value, or proposed decision threshold. The central infant bistability, arbitrary-code learning, access-understanding, trial-yield, and neurophysiological assumptions have not been validated by the source records available to this manuscript. The proposed procedures would require staged feasibility studies, independent scientific review, prospective registration, ethics approval, parental consent, and continuing respect for infant assent or dissent before implementation. Whether the final burden and noncontingent conditions qualify as minimal risk is an ethics determination, not an established property of the proposal.

Abstract

How an infant's percept-related state becomes usable by another person is distinct from whether the infant has phenomenal experience and from whether an observer can decode infant physiology. We define **portable recipient-sensitive relay** as a demanding form of report-like public availability in which an independently estimated state is conveyed through more than one arbitrary action code, remains state-specific under ambiguous stimulation, changes an otherwise uninformed recipient's choice, and is regulated by whether the recipient already has target-resolving access. These criteria are proposed as sufficient operational criteria for this special relay construct, not as necessary conditions for report in general.

The revised design replaces a three-arm epistemic-versus-instrumental comparison with a socially mediated factorial training experiment. During coherent-motion training, recipient target access and current-signal contingency are independently manipulated. In the critical comparison, the same recipient receives the infant's current touch symbol, visibly follows the same code, chooses after the same delay, and produces the same scripted affective and terminal events. The only intended difference is whether the recipient already has target-resolving visual access. Two fixed-schedule noncontingent arms estimate whether any access effect depends on current-signal contingency. A separately randomized direct-instrumental arm remains as a benchmark, but it is no longer used to identify uncertainty reduction because human social mediation necessarily differs from direct computer control. Live yoking is abandoned in favor of fixed, pregenerated schedules, removing dependence of one infant's treatment exposure on another infant's behavior.

At endpoint, optokinetic and electroencephalographic measurements acquired before code revelation would estimate a candidate-report-independent, percept-related latent state. The estimator would not be treated as valid until a separate pilot establishes infant-compatible ambiguity, held-out coherent-to-ambiguous

transport, recurrent direction-related states under competing state models, measurement invariance, and electroencephalographic information incremental to ocular features. Portable state transmission would be tested on recipient-no-access bistable trials, whereas recipient-access suppression would be tested primarily on coherent trials for which access resolves the target. Shared-view bistable trials provide a diagnostic prediction: because seeing the same ambiguous display does not reveal the infant's state, a genuine target-need-sensitive policy should not be generally suppressed there. The primary analysis uses separate group-level criteria for bistable cross-code transmission, coherent redundancy sensitivity, and randomized message efficacy. It does not multiply a behavioral index by an individual transmission gate. Omission is analyzed separately and does not count toward the primary active-symbol information measure.

The original 0.10-bit superiority margin is retained as a provisional scientifically meaningful target, but the earlier enrollment target is withdrawn because it could not support a 0.95 posterior decision rule under its own assumptions. A confirmatory sample size would be chosen only after independent pilots provide decoder reliability, response rates, variance, missingness, and trial-yield estimates and after simulation of the complete multilevel model. Neural source, connectivity, fNIRS, and MRI analyses are exploratory. Even if the behavioral conjunction is supported, the result would establish flexible, recipient-sensitive message efficacy—not phenomenal consciousness, communicative intention, explicit belief attribution, or an infant concept of uncertainty. Failure would constrain the preregistered relay mechanism and effect-size prediction without establishing absence of experience.

Keywords: infancy; nonlinguistic communication; recipient access; social contingency; bistable perception; optokinetic nystagmus; EEG; message passage; causal inference; consciousness

1. Revision Status and Evidential Scope

1.1 What has changed

Independent review identified four defects that prevented the original proposal from serving as a confirmatory causal test.

First, the original training conditions did not isolate recipient epistemic access. A socially mediated condition was compared with a direct computer-control condition and a noncontingent social condition. Those contrasts changed social agency, human causal involvement, current contingency, and event order along with any change in uncertainty reduction. The revised experiment therefore makes recipient target access and current-signal contingency independent factors within the same socially mediated procedure. The same actor, code, current message, visible choice, delay, affect, and terminal event are retained across the critical target-access contrast.

Second, the original endpoint pooled coherent and bistable trials into one access-by-code information subtraction. That was conceptually mistaken. Viewing a coherent display can resolve the objective target; viewing the same bistable display does not reveal the infant's current latent state. The revised endpoint separates three questions:

1. whether an infant's bistable percept-related state is portably conveyed to a recipient without target access;
2. whether signaling is reduced on coherent trials when the recipient already has target-resolving access; and

3. whether randomized passage of an active symbol improves recipient choice.

Shared-view bistable trials are no longer penalized as though the recipient knew the infant's state. They instead test whether infants generalize a simple line-of-sight rule beyond its epistemic relevance.

Third, the original scalar $R^* = D \times R$ could improve when transmission failed, because a negative R became zero whenever the binary gate failed. That composite has been removed. Confirmation now requires separate group-level posterior criteria. Recipient efficacy, portability, and audience regulation cannot compensate for one another.

Fourth, the original physiological and neural program assumed rather than established infant bistability, decoder transport, rapid gaze-code acquisition, and measurement reliability. The randomized trial is now conditional on a staged validation program. If the latent state cannot be validated, the main endpoint is undefined rather than negative.

1.2 What remains unresolved

The redesign improves identification but does not transform recipient-sensitive behavior into direct evidence that infants explicitly represent another person's uncertainty. A binary target-access manipulation can support at least three interpretations:

- sensitivity to the recipient's objective information state;
- learning of an access-conditioned social policy; or
- learning of perceptual cues associated with open and closed access displays.

Novel access geometries, partial-information probes, and shared-view bistable trials can separate some versions of these explanations, but no feasible version of the present experiment uniquely identifies propositional belief attribution, metarepresentation, or a consciously represented uncertainty concept.

The source corpus also remains evidentially distant from several core procedures. It contains broad work on frequency-sensitive learning, methodological records about yoking, computational and developmental work concerning optokinetic nystagmus, computational analogies concerning multimodal representation, technical cautions about EEG–fNIRS acquisition, and philosophical discussions of consciousness and emergence. It does not directly validate the proposed infant ambiguous-motion stimulus, the state decoder, the gaze code, the access geometry, or the trial burden. Those are pilot questions, not facts inferred from peripheral citations.

1.3 Status of the present document

This manuscript is therefore a **staged research blueprint**, not a registration-ready confirmatory protocol. The behavioral causal architecture, estimands, and falsification logic are specified in detail. Final decoder thresholds, demonstration count, endpoint trial count, sample size, and confirmatory priors must be frozen after independent feasibility work and before any confirmatory infant is enrolled.

2. Construct, Research Question, and Boundary Conditions

2.1 Primary research question

The revised primary question is:

Does training in which an infant's current arbitrary signal is socially used when a recipient lacks target-resolving access produce more portable, recipient-sensitive relay than otherwise matched training in which the recipient already has access?

A secondary factorial question is:

Does any effect of recipient target access depend on the recipient's current choice actually being contingent on the infant's current signal?

The design does not ask when consciousness first appears. It asks whether a particular social learning history changes how a percept-related state is translated into public action.

2.2 Portable recipient-sensitive relay

The target construct is renamed **portable recipient-sensitive relay**, abbreviated **PRSR**. It is defined by the following conjunction:

1. **Candidate-report-independent state estimation.** A trial-specific percept-related state is estimated without using the candidate touch or gaze message, recipient choice, reward, or post-response behavior.
2. **Bistable state specificity.** On objectively ambiguous trials, active symbol identity depends on the estimated state rather than only on physical stimulus direction.
3. **Cross-code portability.** State-specific active signaling occurs through both a practiced touch code and an independently mapped, low-exposure gaze code.
4. **Recipient-need sensitivity.** On coherent trials, active information-bearing signaling is reduced when the recipient already has target-resolving access.
5. **No inappropriate ambiguous-access suppression.** Merely seeing the same bistable display does not generally suppress relay, because shared display access does not reveal the infant's latent state.
6. **Message efficacy.** Randomized passage rather than corruption of the infant's active symbol improves the recipient's state-matching choice.
7. **Explanatory controls.** The pattern is not adequately accounted for by first-order discrimination, motor asymmetry, state imbalance, reward exposure, code practice, trial attrition, or a simpler cue-conditioned learning model.

These are not proposed as necessary conditions for ordinary report. A person may report redundantly to a knowledgeable listener, report through only one practiced effector, or report unsuccessfully because the listener ignores the message. PRSR is deliberately narrower. It is an operationally stringent subtype of successful audience-designed signaling relevant to report-like public availability.

2.3 "Private" and "public" without metaphysical inflation

The state is **private relative to the recipient's information path**, not inaccessible to all third-person observation. The recipient cannot directly obtain the target state on no-access trials, whereas the research apparatus may probabilistically estimate it from eye movements and EEG.

The state becomes **publicly usable** when an arbitrary message changes the recipient's choice. This public-use relation does not establish that the latent variable exhausts the infant's percept, that the infant introspectively judges the state, or that the state is phenomenal.

2.4 Percept-related state rather than “the infant's percept”

The latent variable is described throughout as a **percept-related sensorimotor state**. That terminology is intentionally weaker than “the infant's current percept.” A coherent-trained classifier can learn retinal-motion responses, ocular dynamics, stimulus-locked EEG, or other sensorimotor features. Only successful validation across ambiguous input, competing models, visits, modalities, and conditions would justify a more perceptual interpretation. Even then, phenomenal character would remain unmeasured.

2.5 Report-like relay is not consciousness

Successful PRSR would support flexible public availability. It would not prove phenomenal consciousness. A theoretical proposal has argued that phenomenal consciousness may be shared with many animals and may have an adaptive labeling function, but that is an argued evolutionary account rather than a result on which the present experiment depends ^[12].

The evidential asymmetry is central:

- successful relay can establish a flexible causal relation among a state, action code, and recipient;
- failed relay may reflect perceptual, memory, motor, mapping, motivational, social, or measurement limitations and therefore does not establish absent experience.

2.6 No inference of communicative intention from efficacy alone

Message passage can show that a symbol was efficacious. It cannot, by itself, show that the infant intended the recipient to use it. Audience regulation, cross-code portability, and training history make an intentional interpretation more plausible than efficacy alone, but they still do not uniquely establish intention. Terms such as “trying to tell,” “deliberately informing,” and “representing ignorance” are therefore avoided in the confirmatory claims.

3. Evidential Background

3.1 Frequency, contingency, and strong associative alternatives

Frequency-sensitive learning is a serious alternative to any epistemic account. Broad work in language processing describes effects of token and type frequency, practice, cue competition, exemplar learning, and frequency-biased abstraction across multiple domains ^[1]. Related emergentist work emphasizes cue salience and the contingency of form–function associations rather than raw count alone ^[2].

These sources concern language, especially second-language learning. They do not establish how preverbal infants acquire a nonlinguistic relay. Their relevance is methodological: a fair alternative model must include more than a scalar count of exposures. It must allow at least:

- state-specific action counts;
- token and type frequency;

- recency;
- cue salience;
- state–symbol contingency;
- action–outcome contingency;
- cue conjunctions;
- actor and access cues;
- motor costs;
- omission;
- and generalization between code features.

The revised model comparison therefore does not pit a rich epistemic model against a state-free frequency accumulator. All candidate models generate the same trial-level action variables under a common motor and missingness architecture.

3.2 Yoked control and why live yoking is abandoned

Yoked-control procedures are intended to assess contingency by exposing a control participant to a sequence generated independently of that participant's current response ^{[22][20]}. The available records identify the methodological topic but do not provide enough procedural detail to validate every implementation claim.

Live yoking also creates a formal complication: the control infant's treatment exposure depends on another infant's behavior. Potential outcomes then depend on a treatment-assignment vector and yoking policy, not only on the control infant's assigned label. Merely declaring partial interference does not remove that dependence.

The revised design uses **fixed, pregenerated schedules** derived before confirmatory enrollment. Schedule-controlled arms preserve noncontingency without tying one infant's exposure to another infant's realized behavior. Schedule distributions can be calibrated in an independent pilot, but no confirmatory infant determines another confirmatory infant's treatment.

3.3 Social contingency is a construct, not an assumed infant preference

The developmental importance of social contingency has been posed directly in the infant literature ^[23]. The stored record does not establish a settled result, so the manuscript does not assume that social contingency is uniformly preferred, rewarding, or developmentally privileged. The factorial design tests its role under the proposed procedure.

3.4 Optokinetic nystagmus as a candidate-report-independent observation

Optokinetic nystagmus, or **OKN**, is an involuntary oculomotor pattern involved in stabilizing retinal images during relative motion ^[32]. It is an overt action. It is therefore incorrect to call an OKN-based estimator action-independent or behavior-free. The appropriate claim is that OKN is **independent of the candidate touch or gaze report**.

Developmental work has considered OKN in human and monkey infants, although the available title-level record does not establish the reliability of the present use ^[30]. A computational model relates developing motion tuning, binocular vision, vergence, and OKN and simulates perturbations such as strabismus ^[32]. That work supports caution about developmental sensorimotor dependencies; it does not validate trial-specific infant percept decoding.

OKN can also be associated with nonveridical spatial localization. One study reports direction- and phase-dependent auditory mislocalization during reflexive eye movements ^[31]. The source record does not support an age qualifier, so none is added here. The methodological implication is that OKN is not a transparent readout of spatial experience.

3.5 Why coherent calibration is insufficient

A classifier trained on coherent leftward and rightward motion can succeed by detecting physical stimulus direction, retinal motion, ocular artifacts, or stimulus-locked neural responses. Applying its output to ambiguous trials does not automatically make the output perceptual.

The revised validation program therefore requires more than coherent classification. It must compare:

- one-state and two-state models;
- binary and continuous latent-state models;
- independent-trial and temporally persistent models;
- within-trial switching and single-state-per-trial models;
- OKN-only, EEG-only, and combined observation models;
- causal filtering and full-sequence smoothing;
- and invariant versus condition-specific measurement models.

A decoder is acceptable only if its ambiguous-trial interpretation survives these comparisons and if EEG contributes information beyond ocular and eye-movement features.

3.6 Multimodal representation as analogy, not developmental evidence

Computational proposals concerning multimodal internal representation describe coordinated spaces and modality-specific encoders and decoders ^[10]. Goal-oriented semantic signal-processing work describes formal representations and goal-dependent filtering of signal content ^[19]. These frameworks provide structural analogies for translating common information into different output formats. They do not demonstrate infant code portability or consciousness.

3.7 Neuroimaging caution

Simultaneous EEG–fNIRS recording can be technically difficult because the sensors compete for scalp locations ^[38]. Modality addition also need not improve prediction. In one imagined-handwriting benchmark, fNIRS-only performance was near chance and adding fNIRS to EEG changed accuracy only minimally ^[37]. That benchmark does not generalize to infancy or the present task. It supports only the narrow caution that each modality must demonstrate incremental value.

The revised protocol therefore uses EEG and eye tracking together during the primary endpoint and places fNIRS in a separate exploratory block. It does not claim simultaneous EEG–fNIRS coupling or use fNIRS to infer rapid perceptual switching.

4. Staged Research Program

4.1 Rationale for staging

The randomized trial depends on a chain of unvalidated prerequisites:

1. the ambiguous display must generate recoverable direction-related dynamics in late infancy;
2. those dynamics must be measured reliably without candidate report;
3. the coherent-trained measurement system must transport to ambiguous trials;
4. the recipient-access geometry must be understood well enough to influence infant behavior;
5. the independently mapped gaze code must be usable after limited exposure;
6. the trial burden must produce sufficient valid data;
7. randomized message corruption must yield enough active-message trials;
8. the complete decision rule must have acceptable operating characteristics.

If any early link fails, increasing the confirmatory sample does not solve the problem. The program is therefore divided into validation, feasibility, and confirmatory stages.

4.2 Stage 0A: percept and measurement validation

Stage 0A uses an action-free protocol. Infants view coherent and candidate ambiguous displays while eye tracking and EEG are recorded. No touch or gaze symbols are presented, and no recipient is present.

The pilot must establish:

- held-out coherent-direction discrimination;
- recurrent direction-related dynamics under ambiguous input;
- superiority or practical adequacy of a prespecified latent-state model relative to credible one-state, continuous-state, and stimulus-locked alternatives;
- cross-visit reliability;
- within-trial temporal resolution sufficient to define the later response reference state;
- EEG information incremental to eye-position, OKN, blink, and ocular-artifact features;
- and acceptable calibration and artifact rates.

A provisional coherent criterion is held-out balanced accuracy above 0.60 with an interval excluding 0.50 in the intended age range. This is not enough by itself. Ambiguous-trial validation must also show reproducible state recurrence and model transport. Exact predictive-log-density and reliability margins must be frozen before Stage 0A analysis.

If only physical coherent direction can be decoded, the study may proceed as a coherent-stimulus communication study, but it cannot test private-state-specific bistable relay.

4.3 Stage 0B: code and access validation

A separate pilot tests:

- touch-symbol learning;
- gaze-symbol learning after randomized demonstration doses;
- first-trial and early-trial performance before endpoint learning can accumulate;

- response rates in each state and code;
- understanding of the target-access geometry;
- transfer to a novel recipient and room;
- and the shared-view bistable prediction.

The original proposal assumed that six demonstrations would be sufficient for the gaze mapping. That number is now a testable pilot condition rather than an accepted fact. Demonstration doses such as four, six, and eight may be randomized in the pilot. The confirmatory dose must then be fixed prospectively. If usable gaze performance requires extensive reinforced practice, the endpoint can test a second trained code but not low-frequency portability.

4.4 Stage 0C: integrated feasibility and yield

The integrated pilot combines training, endpoint behavior, eye tracking, and EEG. It estimates:

- retention across eight weeks;
- valid training-event count;
- valid endpoint-trial count;
- active-response probability by code and condition;
- frequency of genuine omissions;
- sensor-failure rate;
- amount of data needed for group-level message-efficacy estimates;
- carryover from message corruption;
- state-distribution imbalance;
- and family burden.

The pilot is not used to declare efficacy. Its participants are excluded from the confirmatory analysis.

4.5 Go/no-go status

The confirmatory trial proceeds only if all of the following are met:

1. the central bistable state is interpretable under more than one defensible measurement specification;
2. the candidate-report-independent decoder is sufficiently reliable for attenuation not to make the proposed effect undetectable;
3. gaze-code mapping succeeds on early, nonreinforced endpoint trials;
4. access understanding transfers beyond one superficial display cue;
5. expected active-message counts support group-level passage-versus-corruption estimates;
6. expected attrition and trial yield are ethically and statistically acceptable;
7. complete simulation demonstrates the target posterior assurance.

If these conditions fail, the research program stops or is narrowed. A failed validation gate is not treated as evidence against infant experience or communication in general.

5. Causal Training Experiment

5.1 Factorial structure

The central social experiment crosses two binary factors:

- **Recipient target access**
- $X = 0$: no target-resolving access;
- $X = 1$: target-resolving access to the coherent display.
- **Current-signal contingency**
- $K = 1$: the recipient’s current registered choice depends on the infant’s current symbol;
- $K = 0$: the recipient’s choice follows a fixed, pregenerated schedule independent of the infant’s current symbol.

This produces four socially mediated arms:

Label	Recipient target access	Current-signal contingency	Descriptive name
NSC	0	1	Needed social contingency
RSC	1	1	Redundant social contingency
NSS	0	0	No-access schedule control
RSS	1	0	Resolving-access schedule control

A fifth randomized arm provides a **direct-instrumental benchmark**, abbreviated DI. It is retained to estimate what direct action–outcome control can produce, not to identify the specific effect of recipient need.

5.2 Why NSC versus RSC is the critical comparison

In both NSC and RSC:

- the same human recipient appears;
- the recipient receives the registered current symbol;
- the recipient visibly follows the same codebook;
- the current infant symbol determines the recipient’s first choice;
- the delay from symbol registration to choice is matched;
- recipient facial affect and vocal affect are scripted;
- and the terminal event is matched.

The intended difference is that the RSC recipient already has target-resolving access to the coherent stimulus, whereas the NSC recipient does not. The message is therefore potentially necessary for target selection in NSC and redundant in RSC.

This contrast supports a narrower causal claim than the original manuscript proposed:

training under a socially mediated, target-needed contingent relation causes a different later signaling policy than training under an otherwise matched socially mediated but target-redundant contingent relation.

It does not, by itself, establish that the infant represents entropy, uncertainty, ignorance, or belief.

5.3 Why the noncontingent arms are needed

An NSC–RSC difference might arise from the access display itself. The schedule-controlled arms estimate whether a similar access difference occurs when the recipient's current choice is independent of the current infant signal.

The factorial interaction is:

$$\Delta_{X \times K} = (\mu_{0,1} - \mu_{1,1}) - (\mu_{0,0} - \mu_{1,0}),$$

where:

- $\mu_{x,k}$ is the mean endpoint outcome under recipient-access level x and contingency level k ;
- $x = 0$ denotes no target access;
- $x = 1$ denotes target-resolving access;
- $k = 1$ denotes current-signal contingency;
- $k = 0$ denotes fixed-schedule noncontingency.

A positive interaction indicates that the effect of recipient access is specifically larger when current infant signals lie on the recipient-choice path.

5.4 Fixed schedules rather than live yoking

For $K = 0$ arms, the recipient's symbol input and first choice follow schedules generated before confirmatory enrollment. Schedules are balanced for:

- target direction;
- recipient choice;
- apparent symbol identity;
- target-choice agreement;
- delay;
- terminal event;
- and block position.

They may be calibrated using independent pilot distributions but are not copied from concurrent confirmatory infants. Consequently, each infant's potential outcome depends on that infant's assignment and a fixed protocol, not on another family's behavior.

5.5 Direct-instrumental benchmark

In DI, the touch symbol directly changes a computer-controlled selection. The recipient remains socially visible and acknowledges the resulting event only after the computer commits to it.

DI necessarily differs from the socially mediated arms in causal agency and event meaning. It is therefore a benchmark for direct instrumental acquisition, not the primary control for uncertainty reduction. If DI produces strong practiced-touch behavior but weak gaze transfer and weak recipient regulation, that pattern is informative. If DI matches NSC on all endpoint components, socially mediated recipient need is unnecessary under the tested conditions.

5.6 Training target and mapping

Training uses coherent motion only. Let $S_{ik} \in \{-1, +1\}$ denote the objective direction on training event k for infant i . Two arbitrary touch symbols are mapped to the two directions. Symbol positions vary, preventing a fixed left-side/right-side motor mapping. Mapping is counterbalanced across infants.

Training therefore concerns an occluded or visible **public target**, not an endogenously varying private percept. The defensible generalization claim is:

a learned hidden-target signaling policy may transfer to a percept-related latent state under ambiguous input.

The experiment cannot claim that training directly taught infants to report bistable percepts.

5.7 Social and temporal control

The recipient interacts with the infant through a controlled one-way audiovisual system. During stimulus presentation and symbol registration, the recipient cannot see the infant's eyes, head direction, anticipatory looking, touch trajectory, or affect. The recipient receives only:

- the target display feed when assigned target access;
- the registered symbol when current-signal contingency is active;
- or the pregenerated schedule token in schedule-controlled arms.

The infant sees a standardized recipient image and the access geometry. The recipient's first choice occurs after a fixed delay distribution. The choice is not accompanied by differential praise or disappointment.

This arrangement reduces, but does not eliminate, differences in social meaning. It also sacrifices some naturalistic reciprocity. That tradeoff is required to prevent OKN and other directional behavior from becoming an uncontrolled message path.

5.8 Recipient-access display

The access manipulation must be visually matched from the infant's perspective. A candidate implementation uses identical electrochromic shutters or matched video windows. In the resolving-access condition, the recipient-side target feed is available; in the no-access condition, it is replaced by a luminance- and motion-matched mask. The infant sees a validated cue indicating whether the recipient has target access.

An access cue can become a conditioned stimulus. The protocol therefore includes:

- novel shutter materials;
- reversed spatial layouts;
- novel recipients;
- and a partial-information probe.

Successful transfer across these conditions would reduce, but not eliminate, the learned-barrier explanation.

5.9 Outcome control

The recipient's first choice is the epistemically relevant outcome. A separate terminal event is pleasant but independent of whether that choice matches the target. This prevents reward deprivation when the infant omits or produces a mismatching symbol.

All arms receive matched terminal-event schedules. The protocol separately records:

- recipient-choice accuracy;
- symbol-choice contingency;
- target-choice agreement;
- delay;
- infant affect;
- and terminal-event exposure.

Because recipient-choice accuracy may differ endogenously between arms, it cannot be declared prospectively matched. It is a consequence of the training policy and is modeled as part of the learning history.

5.10 Provisional training dose

The provisional schedule is eight weekly sessions with two short blocks of eight events, for at most 128 events. The earlier proposal of 192 events is withdrawn pending burden data. Sessions terminate on caregiver request or signs of infant distress, fatigue, or sustained disengagement. Actual exposure is recorded as dose, but the primary analysis remains intention-to-treat.

6. Endpoint Architecture

6.1 Separate endpoint modules

The endpoint no longer estimates one pooled access-by-code index. It contains four modules:

1. **Bistable no-access portability**
2. **Coherent recipient-redundancy sensitivity**
3. **Shared-view bistable diagnostic**
4. **Randomized message efficacy**

Touch and gaze codes are tested in each module where feasible.

6.2 Stimulus classes

Let V_{it} denote stimulus class:

- $V_{it} = C$: coherent motion;
- $V_{it} = B$: bistable or ambiguous motion.

Let D_{it} denote physical direction:

- $D_{it} = -1$: coherent leftward;
- $D_{it} = +1$: coherent rightward;
- $D_{it} = 0$: physically identical ambiguous sequence without an assigned left/right direction.

The pilot must establish that the ambiguous display supports interpretable state dynamics. The label “bistable” is conditional on that result.

6.3 Recipient-access conditions

Endpoint access has three levels:

- $E = N$: recipient has no target access;
- $E = R$: recipient has target-resolving access to a coherent display;
- $E = S$: recipient shares visual access to the ambiguous display but does not thereby know the infant’s state.

The primary coherent comparison is N versus R . The primary bistable relay condition is N . The S condition is diagnostic.

6.4 Response codes

Let $C_{it} \in \{T, G\}$ denote the available code:

- T : touch-symbol code;
- G : gaze-symbol code.

The gaze mapping is independently counterbalanced relative to touch. Symbol locations vary by trial. A touch symbol associated with one direction need not share shape, color, or location with the gaze symbol associated with that direction.

The available code is revealed only after the physiological reference epoch. This prevents code-specific preparation from defining the latent state.

6.5 Trial sequence

A provisional endpoint trial contains:

1. **Access cue:** 1.0–1.5 seconds.
2. **Motion display:** approximately 2 seconds.
3. **Reference interval:** approximately 0.5–0.8 seconds with no response symbols visible.
4. **Code revelation:** touch or gaze symbols appear.
5. **Optional response window:** up to 4 seconds.
6. **Private recipient choice:** not visible to the infant.
7. **Standardized terminal event:** independent of the first choice.

Exact timing is pilot-dependent. The state intended for relay is the final stable direction-related state supported during the prespecified late-stimulus and retention reference epoch. A within-trial model is used because state changes may occur during a two-second ambiguous display.

6.6 Preventing uncontrolled recipient access to infant behavior

During the recipient's first choice, the recipient cannot observe:

- OKN;
- eye direction;
- head direction;
- body orientation;
- touch trajectory;
- response latency;
- affect;
- or experimenter behavior.

Only the registered symbol or its randomized corruption is displayed. A one-way system is essential because $P = 0$ message corruption is uninterpretable if the recipient can infer state from visible eye movements.

6.7 Gaze-code familiarization and early transfer

The number of gaze demonstrations is selected in Stage oB. Six remains the provisional dose, not an established sufficient dose. Demonstrations are nonverbal, balanced across states, and position-jittered.

The primary low-exposure transfer estimate uses the first prespecified gaze trials before infants receive visible outcome feedback about gaze-message correctness. Later gaze trials can inform rapid acquisition but cannot be used as evidence of immediate transfer without a trial-wise learning model.

6.8 Optional signaling and omission

The infant is not required to answer. Let:

- $R_{it} = 1$ if an active symbol is validly selected;
- $R_{it} = 0$ if no active symbol is selected during an otherwise valid response window.

If $R_{it} = 1$, let $K_{it} \in \{1, 2\}$ denote active symbol identity.

A true omission requires:

- functioning response equipment;
- adequate eye-tracking or video visibility;
- completion of the reference epoch;
- and no valid touch or gaze selection.

Sensor failure is missing data, not omission.

Omission is not treated as a primary report symbol. State-dependent omission may arise from fatigue, motor asymmetry, disengagement, or selective response. It is modeled as behavior and reported in secondary ternary-information analyses, but the primary state-content measure depends on active symbol identity and penalizes low active-response rates.

6.9 Provisional endpoint burden

A provisional schedule contains 80 response trials over four visits, supplemented by short action-free calibration blocks. One possible allocation is:

- 16 early bistable no-access transfer trials;
- 16 additional bistable no-access trials;
- 16 shared-view bistable trials;
- 32 coherent audience-access trials.

Message passage is randomized within eligible no-access trials rather than added as a separate large module. These counts are illustrative. They are not frozen until the integrated pilot establishes usable-trial yield and burden.

7. Candidate-Report-Independent State Measurement

7.1 Latent process

Let j index short time bins within trial t . Let $Z_{itj} \in \{-1, +1\}$ denote a direction-related latent state. For the first usable bin:

$$\Pr(Z_{it1} = z) = \sigma \left[z \left(\alpha_i + b_D D_{it} + \mathbf{b}_u^\top \mathbf{u}_{it1} \right) \right],$$

where:

- $z \in \{-1, +1\}$;
- α_i is an infant-specific directional bias;
- b_D is the coherent physical-direction effect;
- $\mathbf{u}_{it1}$ contains measurement nuisance variables;
- $\sigma(x) = 1/(1 + \exp[-x])$.

For later bins:

$$\Pr(Z_{itj} = z \mid Z_{it,j-1}) = \sigma \left[z \left(\alpha_i + b_D D_{it} + \phi_i Z_{it,j-1} + \mathbf{b}_u^\top \mathbf{u}_{itj} \right) \right].$$

This formulation avoids assigning 0 to a variable declared to take only -1 or $+1$. It also permits within-trial switching.

The response-reference state Z_{it}^{ref} is defined from the posterior state occupancy during a prespecified late-stimulus and retention interval. If no state dominates that interval above a pilot-validated stability threshold, the trial contributes probabilistically rather than receiving a hard label.

7.2 Observation model

Let:

- $\mathbf{O}_{itj}$ be OKN and eye-movement features;

- $\mathbf{H}_{itj}$ be EEG features;
- $\mathbf{W}_{itj} = (\mathbf{O}_{itj}, \mathbf{H}_{itj})$.

A robust observation model is:

$$\mathbf{W}_{itj} \mid Z_{itj} = z, \mathbf{u}_{itj} \sim t_{\nu}(\boldsymbol{\mu}_z + \mathbf{B}\mathbf{u}_{itj}, \boldsymbol{\Sigma}_z).$$

The nuisance vector includes:

- absolute gaze position;
- pupil visibility;
- slow-phase velocity magnitude;
- blink proximity;
- head-motion estimate;
- EEG artifact burden;
- alertness estimate;
- stimulus luminance;
- visit;
- and elapsed session time.

Recipient access, code, group, and visit are not simply omitted and assumed irrelevant. Measurement invariance is explicitly tested. A model with condition-specific nuisance shifts is compared with a shared model. The state score must not function as a classifier of treatment arm, access cue, code, or visit after physical direction and quality are controlled.

7.3 Filtering rather than future-dependent smoothing

The primary state posterior uses only physiology available up to the end of the response-reference interval:

$$\pi_{it}(z) = \Pr(Z_{it}^{\text{ref}} = z \mid \mathbf{W}_{i,\leq t, \leq \text{ref}}, \mathbf{D}_{i,\leq t}, \boldsymbol{\Theta}_W).$$

Future trials and post-response physiology do not update the primary state estimate. Full-sequence smoothing is reported only as a sensitivity analysis. Material disagreement between filtering and smoothing is a measurement warning, not an opportunity to choose the more favorable result.

7.4 Excluded variables

The primary measurement posterior excludes:

- active response indicator R_{it} ;
- symbol identity K_{it} ;
- touch trajectory;
- gaze-symbol fixation after code revelation;
- recipient choice;
- message-passage assignment;
- terminal event;
- training group label;

- and post-response physiology.

Access cues occur before the reference interval and can alter arousal or attention. Excluding the access label from decoder inputs does not remove that contamination. The validation therefore includes residual condition classification and condition-specific calibration tests.

7.5 Incremental EEG criterion

EEG may contain ocular artifact. Cross-modal agreement is not independent convergence if EEG merely reconstructs eye movement. The pilot must compare:

1. an eye/OKN model;
2. an EEG model after ocular-feature regression and artifact control;
3. a combined model;
4. a combined model with all eye and ocular-artifact features available.

EEG counts as incremental only if it improves held-out prediction or latent-state stability beyond the richest ocular model and if that improvement replicates across infants or visits. If it does not, the state estimator may remain OKN-based, but claims of multimodal convergence are withdrawn.

7.6 Competing latent-state models

The pilot comparison includes at least:

- one-state observation model;
- two-state hidden Markov model;
- continuous latent-direction model;
- independent-bin mixture model;
- independent-trial binary model;
- within-trial switching model;
- coherent-trained discriminative decoder;
- and a stimulus-locked model without endogenous bistable dynamics.

A binary model is not accepted merely because it generates two states. Acceptance depends on held-out prediction, recurrence, stability, and transport.

7.7 Modular uncertainty propagation

The analysis uses a frozen measurement posterior:

$$p_{\text{cut}}(\mathbf{Z}, \Theta_W \mid \mathbf{W}, \mathbf{D}).$$

Draws from this posterior are propagated into the behavioral model, but endpoint actions do not update the state measurement. This is a modular cut, not a fully coherent joint Bayesian posterior. Uncertainty is propagated **modularly**, not jointly in the ordinary Bayesian sense.

8. Behavioral Observation Model and Information Measures

8.1 Separating response propensity from symbol identity

The primary behavioral model has two parts.

First, active-response propensity:

$$R_{it} \sim \text{Bernoulli}(\rho_{it}),$$

$$\text{logit}(\rho_{it}) = \eta_i + \eta_Z Z_{it}^{\text{ref}} + \boldsymbol{\eta}_X^\top \mathbf{x}_{it},$$

where $\mathbf{x}_{it}$ includes stimulus class, recipient access, code, training arm, trial order, motor quality, and prespecified interactions.

Second, active symbol identity:

$$K_{it} \mid R_{it} = 1 \sim \text{Bernoulli}(\theta_{it}),$$

where the two active symbols are coded 0 and 1, and

$$\text{logit}(\theta_{it}) = \beta_i + \beta_Z Z_{it}^{\text{ref}} + \boldsymbol{\beta}_X^\top \mathbf{x}_{it}.$$

This decomposition prevents a state-dependent omission pattern from masquerading as accurate active-symbol coding.

8.2 Standardized symbol information

Raw mutual information depends on state entropy. If one condition contains mostly one latent state, its information ceiling is lower. The primary sensitivity measure therefore standardizes the latent-state distribution to 1/2 for each state.

For a specified group and endpoint condition q , define:

$$I_{1/2}^q = \sum_{z \in \{-1, +1\}} \frac{1}{2} \sum_{k \in \{1, 2\}} p(k \mid R = 1, z, q) \log_2 \left[\frac{p(k \mid R = 1, z, q)}{\frac{1}{2}p(k \mid R = 1, -1, q) + \frac{1}{2}p(k \mid R = 1, +1, q)} \right].$$

This is the information in active symbol identity under a balanced latent-state distribution.

8.3 Effective active-symbol information

A conditional active-symbol channel can look accurate even if active responses are extremely rare. Define:

$$\rho_{\min}^q = \min_{z \in \{-1, +1\}} \Pr(R = 1 \mid Z = z, q),$$

and:

$$J^q = \rho_{\min}^q I_{1/2}^q.$$

The quantity J^q is an **effective active-symbol information score**, measured in bit-weighted trials. It is not asserted to be literal realized mutual information. Its properties are useful for the present purpose:

- it is zero if one state never receives an active response;
- it cannot improve merely because omissions become more frequent;
- it excludes omission as a communicative token;
- and it preserves a direct report of the underlying symbol channel $I_{1/2}^q$.

All analyses report ρ_{-1} , ρ_{+1} , $I_{1/2}$, and J separately. No conclusion rests on J alone.

8.4 Secondary ternary information

For comparison with broader communication models, a secondary measure treats $A \in \{0, 1, 2\}$, where 0 is omission:

$$I_{1/2}(Z; A \mid q).$$

This measure can reveal state-dependent withholding, but it is not counted as primary report-like transmission unless the recipient demonstrably uses an explicit no-response token. Genuine omission and technical corruption are visually distinct in the revised protocol.

8.5 Bistable portability components

For training group g and code c , define:

$$J_{g,c}^{B,N} = J^q \quad \text{for} \quad q = (G = g, V = B, E = N, C = c).$$

Portable bistable transmission requires positive, practically meaningful $J_{g,T}^{B,N}$ and $J_{g,G}^{B,N}$. The two values remain separate. A minimum-across-codes statistic is not the sole endpoint because extrema are unstable in sparse cells.

The early gaze estimate is restricted to the prespecified initial nonreinforced trials:

$$J_{g,G,\text{early}}^{B,N}.$$

Later gaze performance is analyzed with a trial-wise learning term.

8.6 Coherent redundancy sensitivity

On coherent trials, recipient access resolves the objective target. Define state-standardized active-response probability:

$$\bar{\rho}_{g,c}^{C,e} = \frac{1}{2} \sum_{z \in \{-1, +1\}} \Pr(R = 1 \mid Z = z, G = g, V = C, E = e, C = c).$$

Response suppression is:

$$S_{\rho,g,c} = \bar{\rho}_{g,c}^{C,N} - \bar{\rho}_{g,c}^{C,R}.$$

Information-bearing suppression is:

$$S_{J,g,c} = J_{g,c}^{C,N} - J_{g,c}^{C,R}.$$

Both are reported because low informed-condition information alone could arise from constant, random, or absent signaling. PRSR predicts reduced active signaling and reduced effective active-symbol information when coherent target access makes the message redundant.

8.7 Shared-view bistable diagnostic

For bistable trials:

$$H_{g,c}^B = J_{g,c}^{B,N} - J_{g,c}^{B,S}.$$

A strong line-of-sight heuristic predicts $H_{g,c}^B > 0$: signaling falls whenever the recipient can see the display. A target-need-sensitive policy predicts approximate equivalence because shared ambiguous input does not identify the infant's state.

Equivalence requires a preregistered region:

$$|H_{g,c}^B| < \epsilon_B.$$

The margin ϵ_B must be scientifically justified and pilot-informed before confirmation. Failure of equivalence is not automatically evidence against all recipient sensitivity; it specifically weakens the claim that infants regulate relay according to target-relevant information rather than visible access alone.

9. Randomized Message Efficacy

9.1 Passage versus corruption

Let $P_{it} \in \{1, 0\}$ be randomized before the trial:

- $P = 1$: the infant's active symbol is passed;
- $P = 0$: it is replaced by a perceptually matched corrupted symbol.

Corruption uses either symbol flipping or a balanced independent symbol draw according to a preregistered schedule. It does not produce a blank. Thus, corruption is distinguishable from genuine omission, and the recipient cannot infer passage status from display format.

The recipient is blind to P . The infant does not see the recipient's first choice and therefore receives no immediate evidence that corruption occurred.

9.2 Eligibility and interpretation

The primary passage analysis conditions on a valid active symbol because the randomization occurs before action and passage is enacted only if an active symbol exists. Conditioning on active response does not compromise the randomized comparison between passage and corruption within the active-message stratum, provided passage assignment cannot influence whether the infant acts. That assumption is credible because P is concealed from the infant and apparatus until after registration.

The analysis establishes message efficacy, not communicative intention.

9.3 Balanced recipient accuracy

Let $Q_{it} \in \{-1, +1\}$ be the recipient's target choice. Define balanced accuracy:

$$\text{BA}_{g,c}^p = \frac{1}{2} \Pr(Q = -1 \mid Z = -1, G = g, C = c, P = p) + \frac{1}{2} \Pr(Q = +1 \mid Z = +1, G = g, C = c, P = p)$$

Balanced accuracy has a 0.50 no-information baseline even when latent-state prevalence is imbalanced.

The message-efficacy effect is:

$$\Gamma_{g,c} = \text{BA}_{g,c}^1 - \text{BA}_{g,c}^0.$$

PRSR requires:

$$\Gamma_{g,T} > 0, \quad \Gamma_{g,G} > 0,$$

and:

$$\text{BA}_{g,T}^1 > 0.50, \quad \text{BA}_{g,G}^1 > 0.50.$$

These are group-level quantities. No infant-specific binary gate is constructed.

9.4 Proper-score sensitivity analysis

Recipient confidence, if collected, permits a proper-score analysis. Let $\hat{q}_{it}$ be the recipient's probability assigned to $Z = +1$. A log-score improvement is:

$$\Gamma_{g,c}^{\log} = \mathbb{E}[\log p_{\text{rec}}(Z \mid P = 1) - \log p_{\text{rec}}(Z \mid P = 0)].$$

This tests whether passage improves calibrated belief rather than only binary accuracy. It is secondary because infant-friendly recipient procedures may favor discrete choices.

9.5 Blocked-trial precision

The original proposal attempted infant-specific passage effects from a rare subset divided across two codes. The revised study estimates passage effects at the group level with hierarchical participant effects. The required number of randomized active-message trials is determined by integrated-pilot response rates and simulation. If active-message counts are inadequate, recipient use cannot be a confirmatory component.

10. Causal Estimands and Conjunctive Decision Rule

10.1 Potential outcomes

Let $Y_i(x, k)$ denote an endpoint component infant i would exhibit under recipient-access assignment x and current-contingency assignment k . Fixed schedules eliminate live-yoking interference. Under consistency, allocation integrity, and no cross-family interaction, randomization identifies:

$$\mu_{x,k}^Y = \mathbb{E}[Y_i(x, k)].$$

For each bistable code c , define the access effect under current contingency:

$$\Delta_{\text{need},c} = \mu_{0,1}^{J_c^{B,N}} - \mu_{1,1}^{J_c^{B,N}}.$$

Define the factorial interaction:

$$\Delta_{\text{int},c} = \left(\mu_{0,1}^{J_c^{B,N}} - \mu_{1,1}^{J_c^{B,N}} \right) - \left(\mu_{0,0}^{J_c^{B,N}} - \mu_{1,0}^{J_c^{B,N}} \right).$$

Analogous estimands are defined for S_ρ , S_J , and Γ .

10.2 Primary causal claim

The strongest supported causal statement is conditional:

socially mediated current-signal training under recipient target need produces more later portable and recipient-sensitive relay than otherwise matched socially mediated current-signal training under recipient target redundancy, and this access effect is larger than the corresponding access difference under schedule-controlled noncontingency.

This remains weaker than “infants learned to represent another person’s uncertainty.”

10.3 Provisional component margins

The original proposal used a 0.10-bit superiority margin. The revised proposal retains 0.10 as a provisional target for the controlled bistable active-information contrast:

$$\delta_B = 0.10.$$

Other provisional margins are:

- absolute active-information target $j_0 = 0.05$;
- coherent response-suppression target $\delta_\rho = 0.10$;
- coherent information-suppression target $\delta_J = 0.03$;
- message-efficacy target $\delta_\Gamma = 0.05$ in balanced accuracy;
- shared-bistable equivalence margin ϵ_B , to be fixed from validation and measurement precision.

These values are not empirical estimates. Final margins must be frozen before confirmatory enrollment and cannot be selected to fit pilot effect estimates. Pilot data inform feasibility and measurement error; scientific relevance determines the superiority margin.

10.4 Conjunctive confirmation criteria

The strong PRSR training hypothesis is supported only if the joint posterior satisfies all of the following:

Measurement validity

The preregistered latent-state validation gates are passed. If not, the behavioral state-relay hypothesis is not evaluated.

Absolute bistable transmission in NSC

For both touch and early gaze:

$$J_{\text{NSC},T}^{B,N} \geq j_0,$$

$$J_{\text{NSC},G,\text{early}}^{B,N} \geq j_0.$$

Controlled training effect

For both codes:

$$\Delta_{\text{need},c} \geq \delta_B,$$

and:

$$\Delta_{\text{int},c} > 0.$$

A final registration may assign the 0.10 margin to the interaction rather than both simple contrasts if simulation shows that this better matches the causal question. That choice must precede confirmatory data.

Coherent recipient-redundancy sensitivity

In NSC, both codes show:

$$S_{\rho,\text{NSC},c} \geq \delta_\rho,$$

and:

$$S_{J,\text{NSC},c} \geq \delta_J.$$

Shared-view bistable preservation

For both codes:

$$|H_{\text{NSC},c}^B| < \epsilon_B.$$

Randomized message efficacy

For both codes:

$$\Gamma_{\text{NSC},c} \geq \delta_\Gamma,$$

and passed-message balanced accuracy exceeds 0.50.

The proposed posterior decision rule is:

$$\Pr(\text{all confirmatory component inequalities} \mid \text{data}) \geq 0.95.$$

This rule is intentionally demanding. A partial pattern supports a narrower claim rather than being forced into a binary global interpretation.

10.5 Why no scalar composite is used

No single number can safely combine:

- state specificity;
- response rate;
- active symbol accuracy;
- code portability;
- coherent suppression;
- ambiguous-access preservation;
- and recipient efficacy.

A weighted sum would allow one component to compensate for another. A binary multiplier would create nonmonotonic scores. The revised analysis therefore treats PRSR as a multivariate conjunction and publishes every component.

11. Comparable Learning Models

11.1 Common observational architecture

All candidate models generate the same observed variables:

1. whether the infant makes an active response;
2. which active symbol is selected;
3. response latency;
4. recipient choice as determined by the experimental rule;
5. terminal event;
6. weekly probe behavior;
7. endpoint behavior.

They share the same:

- state or target inputs;
- motor-bias terms;
- code mapping;
- recipient identity effects;
- trial-order effects;
- omission model;
- missingness model;
- and hierarchical participant structure.

The models differ only in learning-state representations and update rules.

11.2 Shared action policy

For training event k , let $s_{ik} \in \{-1, +1\}$ be the coherent target and $a \in \{0, 1, 2\}$ an omission or active symbol. Each model produces action utilities:

$$U_{ik}(a) = m_i(a) + \mathbf{b}^\top \mathbf{x}_{ik}(a) + L_{ik}^{(m)}(a),$$

where:

- $m_i(a)$ captures infant-specific motor preference;
- $\mathbf{x}_{ik}(a)$ contains common perceptual and contextual predictors;
- $L_{ik}^{(m)}(a)$ is the learning contribution under model m .

Action probability is:

$$\Pr(A_{ik} = a) = \frac{\exp[U_{ik}(a)]}{\sum_{a'} \exp[U_{ik}(a')]}.$$

Every model can therefore represent a directional code. None is structurally forced to choose one globally preferred action.

11.3 Rich associative-frequency model

The associative model tracks state-, symbol-, cue-, and outcome-specific counts:

$$\mathbf{F}_{i,k+1} = \lambda_F \mathbf{F}_{ik} + \mathbf{f}(s_{ik}, a_{ik}, x_{ik}, k_{ik}, y_{ik}),$$

where:

- $0 \leq \lambda_F \leq 1$ implements recency;
- $\mathbf{f}$ records token counts, type counts, target–symbol co-occurrence, access-cue conjunctions, contingency cues, actor identity, and outcome occurrence.

Its learning contribution is:

$$L_{ik}^{(F)}(a) = \beta_F^\top \mathbf{F}_{ik}(s_{ik}, a).$$

This is a strong cue-conditioned learner. It can acquire barrier-specific and actor-specific policies and can exploit cue competition. It is not expected to fail merely because prospectively delivered exposure totals are matched.

11.4 State-conditioned reinforcement model

The instrumental model has state- and context-conditioned action values:

$$Q_{i,k+1}(s, x, a) = Q_{ik}(s, x, a) + \lambda_i 1(s_{ik} = s, x_{ik} = x, A_{ik} = a) [r_{ik} - Q_{ik}(s, x, a)].$$

Its learning contribution is:

$$L_{ik}^{(Q)}(a) = \beta_{Q,i} Q_{ik}(s_{ik}, x_{ik}, a).$$

The context x may include access, actor, and code. Reward r_{ik} may be decomposed into:

- terminal-event value;
- recipient-choice agreement;
- social-affect value;
- and action cost.

This model can learn the directional code and access-dependent policies if those policies are reinforced. It is therefore a credible competitor rather than a state-free straw model.

11.5 Access-rule model

The access-rule model adds a learned value for signaling under observable access cues:

$$L_{ik}^{(A)}(a) = L_{ik}^{(Q)}(a) + \beta_{A,i} 1(a \neq 0) 1(X_{ik} = 0) 1(K_{ik} = 1).$$

It represents “signal under this access configuration” without representing recipient uncertainty. It may generalize across perceptually similar barriers but need not scale with graded uncertainty or survive novel implementations.

11.6 Objective information-gain model

Let S be the training target and V_R the information objectively available to the recipient before the message. Define recipient target entropy:

$$H_R = H(S \mid V_R).$$

For candidate active message a , define expected target entropy after the message:

$$H_R^{\text{post}}(a) = \mathbb{E}[H(S \mid V_R, M = a)].$$

Objective information gain is:

$$\text{IG}_{ik}(a) = H_R - H_R^{\text{post}}(a).$$

The learning contribution is:

$$L_{ik}^{(\text{IG})}(a) = L_{ik}^{(Q)}(a) + \beta_{\text{IG},i} \text{IG}_{ik}(a) 1(K_{ik} = 1).$$

In no-access coherent trials, prior target entropy is high and a valid message can reduce it. In resolving-access trials, target entropy is already low and information gain is near zero.

This model operationalizes the experimenter-defined information structure. It does not imply that the infant explicitly computes Shannon entropy.

11.7 Identifiability limits

A flexible access-rule model may approximate the information-gain model in a binary access task. To improve discrimination, held-out probes include:

- partial recipient information;

- novel access geometries;
- recipient access through a different sensory route;
- and shared-view bistable trials.

Even these probes cannot prove an explicit uncertainty representation. Model comparison supports one formal predictive organization relative to specified alternatives; it does not reveal the unique psychological algorithm.

11.8 Model comparison

Models are compared using:

- held-out predictive log density;
- calibration of response and omission probabilities;
- prediction of early gaze transfer;
- prediction under novel access configurations;
- prediction of coherent suppression;
- prediction of shared-view bistable preservation;
- and posterior predictive checks.

Parameter counts and priors are matched or regularized to avoid rewarding the most flexible model automatically. Model recovery is tested in simulation before infant data are interpreted.

12. Sample Size and Design Assurance

12.1 Withdrawal of the original sample claim

The earlier target of 72 analyzable infants per group was inconsistent with a 0.95 posterior probability that both effects exceeded a 0.10-bit margin under the stated mean assumptions.

For two independent groups with standard deviation $\sigma = 0.18$ bits and n infants per group, the approximate standard error is:

$$SE = \sqrt{\frac{2\sigma^2}{n}}.$$

At $n = 72$:

$$SE \approx \sqrt{\frac{2(0.18)^2}{72}} \approx 0.030 \text{ bits.}$$

If the true effect is 0.11 bits, an estimate centered at 0.11 with standard error 0.03 has only modest probability of exceeding 0.10. Requiring several component inequalities jointly makes assurance lower still.

No final enrollment target is therefore asserted.

12.2 Illustrative normal-approximation scenarios

For a one-sided probability near 0.95 that an effect exceeds margin $m = 0.10$, a rough per-arm requirement is:

$$n \approx 2\sigma^2 \left[\frac{1.645}{\delta - m} \right]^2,$$

where δ is the assumed true effect.

Using $\sigma = 0.18$:

Assumed true effect δ	Distance above 0.10 margin	Approximate n per compared arm
0.11	0.01	1,754
0.13	0.03	195
0.15	0.05	70
0.18	0.08	28

These values are illustrative and omit latent-state error, clustering across trials, missingness, multiplicity, factorial interactions, and the conjunctive decision rule. They show why powering around an expected effect barely above the superiority margin is untenable.

12.3 Simulation requirements

Final assurance simulation must include:

- all five randomized arms;
- the access-by-contingency interaction;
- repeated trials nested within infant;
- decoder sensitivity and specificity;
- posterior uncertainty in Z ;
- state imbalance;
- state-dependent omission;
- early versus late gaze learning;
- active-message counts;
- passage-versus-corruption randomization;
- differential attrition;
- visit effects;
- missing-not-at-random sensitivity;
- and the joint posterior decision rule.

Operating characteristics must be reported under:

- the null;
- partial-component alternatives;

- the target alternative;
- decoder degradation;
- access-cue confounding;
- and model misspecification.

12.4 Sample selection after pilot

The confirmatory sample size may use pilot estimates of variance and usable-trial rate, but not an inflated pilot efficacy estimate. Conservative effect scenarios should be prespecified. If the necessary sample is infeasible, the appropriate response is to lower the claim, change the margin for scientific reasons, simplify the decision rule, or abandon confirmation—not to present an underpowered study as decisive.

13. Language Development and Longitudinal Scope

13.1 Enrollment and measurement

Infants are provisionally enrolled near 9 months and followed into late infancy. The protocol does not assume that all infants at a given chronological age are before productive word combinations. Productive language is measured with:

- weekly caregiver diaries;
- recorded naturalistic samples;
- masked independent coding;
- a vocabulary inventory;
- and explicit rules for formulaic routines and multilingual combinations.

13.2 No full-relay change-point claim from touch probes

Weekly touch probes cannot establish onset of the full PRSR construct because they do not repeatedly measure:

- bistable specificity;
- gaze portability;
- coherent recipient suppression;
- shared-view preservation;
- or randomized message efficacy.

The original claim that a weekly change point marked onset of full relay is withdrawn. Weekly data estimate only the acquisition trajectory of the practiced touch policy.

13.3 Relation to word combinations

At endpoint, infants are classified by whether consistent productive word combinations have been observed. Because onset is interval-censored and post-randomization, language status is not used to exclude infants from the primary intention-to-treat analysis.

Secondary analyses estimate:

- PRSR components among infants not yet observed to combine words;
- interactions between training and language status;
- and the interval relation between touch-policy change and language onset.

The study can show PRSR in infants who have not yet met the operational word-combination criterion. It cannot establish the precise developmental onset of PRSR unless the full conjunction is assessed longitudinally.

14. Neural and Physiological Analyses

14.1 Status as exploratory work

The neural component is exploratory until pilot reliability is established. The source records do not directly support the proposed infant posterior-parietal, medial-prefrontal, pulvinar, or mediodorsal mechanisms. The manuscript therefore does not designate a confirmatory infant relay circuit.

14.2 EEG analyses

Dense-array EEG is acquired during the candidate-report-independent window. Exploratory questions include:

- whether direction-related patterns generalize across trials later assigned to touch and gaze;
- whether such generalization remains after controlling physical coherent direction;
- whether validated bistable states have separable EEG patterns;
- and whether recipient-access cues alter pre-response dynamics.

Code-invariant decoding on coherent trials is not sufficient because physical direction is shared across codes. The stronger test uses validated bistable states and controls for ocular features.

14.3 Source and connectivity limitations

Exploratory source models may compare posterior and frontal temporal organization. Any directional-connectivity estimate is model-dependent and vulnerable to:

- common input;
- volume conduction;
- source leakage;
- reference choice;
- head-model uncertainty;
- and access-related arousal.

Work on nonverbal dyads has used relevant-interval selection and Granger analysis to infer directional temporal predictability ^[36]. Such methods do not establish interventionist biological causation. The present manuscript therefore uses “directional predictability” rather than “causal neural influence.”

14.4 fNIRS protocol

fNIRS is collected in a separate exploratory session with eye tracking but without simultaneous dense EEG. This fixed scheme avoids alternating ambiguously between integrated and separate acquisition.

fNIRS is used for block-level superficial cortical responses to recipient access and social contingency. It is not used to:

- decode rapid perceptual switches;
- measure the thalamus;
- infer deep-brain activity;
- or estimate trial-level EEG–fNIRS effective connectivity.

14.5 MRI

Optional natural-sleep structural and diffusion MRI is exploratory. Any relation between tractography and behavior is correlational and potentially selected by sleep-scan completion. Named thalamocortical analyses are hypothesis-generating. Diffusion metrics do not establish task-level signaling through a tract.

14.6 No causal mediation claim

Neural measurements and endpoint behavior are obtained within the same developmental period. Randomization can identify effects of training on a neural estimate, but it does not randomize the neural estimate or establish mediator-before-outcome order. The revised manuscript therefore removes confirmatory causal mediation.

At most, the study can test **associational mechanistic consistency**:

- training affects a neural measure;
- the neural measure covaries with a behavioral component;
- and the pattern is temporally compatible with the proposed sequence.

That conjunction does not prove that the neural measure mediates the behavioral effect.

15. Alternative Explanations and Diagnostic Patterns

15.1 Rich associative learning

Explanation: Infants learn a conjunction of target, barrier, actor, and symbol cues without representing recipient need.

Diagnostic: Compare rich associative, access-rule, reinforcement, and information-gain models under novel access geometries and partial-information probes.

Remaining limitation: A sufficiently flexible associative learner can approximate many recipient-sensitive patterns. Model superiority is relative, not proof of mental-state representation.

15.2 Direct instrumental learning

Explanation: Action–outcome control alone produces portability.

Diagnostic: The DI benchmark tests whether direct control yields the same bistable cross-code, coherent suppression, and message-efficacy profile.

Interpretation: If DI matches NSC, recipient target need is unnecessary under this task. If DI produces only practiced-touch competence, it supports a dissociation between instrumental control and PRSR.

15.3 Social agency

Explanation: A person on the signal-to-choice path is more motivating than a computer.

Diagnostic: NSC and RSC both place the same person on that path. The central target-access contrast therefore holds social agency constant more closely than the original design.

Remaining limitation: Recipient access may subtly alter social meaning, attention, or perceived authenticity. No laboratory control can guarantee identical subjective salience.

15.4 Current contingency

Explanation: Any current social contingency, not target need, drives learning.

Diagnostic: RSC has current social contingency without target need. The noncontingent arms estimate the access-by-contingency interaction.

Interpretation: A general NSC–RSC null combined with both contingent arms exceeding schedule arms supports social contingency but not recipient need.

15.5 Learned barrier or display cue

Explanation: Infants learn to respond differently to a visible access cue.

Diagnostic: Novel barriers, changed geometry, partial access, and shared-view bistable trials test transfer.

Remaining limitation: Generalization can still be implemented by learned perceptual categories rather than another-agent reasoning.

15.6 Shared ambiguous access

Explanation: Infants suppress whenever the recipient can see any display.

Diagnostic: On shared-view bistable trials, the recipient sees the same objective input but does not know the infant's state. General suppression here favors a line-of-sight heuristic over state-relevant recipient need.

15.7 First-order perceptual improvement

Explanation: Training changes motion discrimination rather than communication.

Diagnostic: Action-free coherent discrimination, state-decoder calibration, OKN gain, and noncommunicative anticipation are measured separately.

Interpretation: A group difference in first-order perception is scientifically relevant but complicates the relay contrast. Baseline adjustment cannot fully repair a treatment-induced perceptual difference.

15.8 Motor asymmetry

Explanation: One state is easier to express because of side preference or code geometry.

Diagnostic: Symbol positions vary, mappings are counterbalanced, motor control trials are included, and state-specific response rates are reported.

15.9 Selective omission

Explanation: Apparent information arises because the infant disengages selectively.

Diagnostic: Primary active-symbol information excludes omission as a symbol and is weighted by the lower state-specific active-response rate. Ternary information and omission models are secondary.

15.10 Endpoint learning of the gaze code

Explanation: Gaze performance is acquired during testing rather than transferred.

Diagnostic: The primary gaze result is restricted to early nonreinforced trials. Later performance is modeled as within-endpoint learning.

15.11 Recipient decoding by uncontrolled behavior

Explanation: The recipient reads OKN, head direction, or affect.

Diagnostic: The recipient sees only the registered or corrupted message during the first choice. Eye, head, body, and latency cues are unavailable.

15.12 State-decoder circularity

Explanation: The action defines the state it allegedly reports.

Diagnostic: Candidate actions and post-response variables are excluded from the frozen measurement posterior.

Remaining limitation: The decoder may still learn report-independent motor or ocular correlates rather than a percept. Pilot construct validation remains essential.

15.13 Binary-state imposition

Explanation: A two-state model manufactures bistability.

Diagnostic: Binary, continuous, one-state, independent, and switching models are compared out of sample. A two-state interpretation is not accepted solely because it fits in sample.

15.14 Linguistic contribution

Explanation: Receptive words or covert labels support performance.

Diagnostic: Direction words and code names are excluded from the task, and productive language is measured.

Remaining limitation: Receptive or covert linguistic contributions cannot be eliminated. The study concerns nonverbal output before or around productive combinations, not a language-free cognitive system.

16. Falsification and Interpretation Rules

16.1 Measurement failure

If Stage 0 does not validate the ambiguous latent state, the confirmatory bistable endpoint is undefined. Strong coherent action–stimulus associations cannot substitute for bistable state specificity.

16.2 Minimum-effect falsification

If the posterior upper bound for a prespecified controlled bistable contrast lies below 0.10 bits, the result falsifies the **preregistered minimum-effect prediction under the fitted model**. It does not falsify every weaker recipient-need effect.

16.3 Portability failure

If practiced touch is state-specific but early gaze is not, the result supports trained-code signaling rather than portable relay. A later gaze increase may support rapid acquisition, not immediate cross-code transfer.

16.4 Audience-regulation failure

If coherent signaling is equally strong under no access and target-resolving access, the strong recipient-sensitive construct is unsupported even if the message is informative.

This does not imply that the behavior is not report in an ordinary sense. Redundant reports remain possible.

16.5 Inappropriate bistable suppression

If signaling is suppressed when the recipient merely shares the ambiguous display, the result favors a broad line-of-sight or access-cue policy over sensitivity to whether the recipient can infer the infant's state.

16.6 Message-efficacy failure

If passage does not improve balanced recipient accuracy relative to corruption, the active behavior may remain state-correlated but does not satisfy the successful-relay criterion.

16.7 Factorial identification failure

If NSC exceeds RSC but an equivalent access difference occurs in schedule-controlled arms, the result does not support a contingency-specific recipient-need effect. It may reflect the access display or general context.

16.8 Direct-instrumental equivalence

If DI and NSC are equivalent across all PRSR components, direct action–outcome learning is sufficient under the tested conditions. The stronger socially mediated target-need mechanism is unnecessary.

16.9 Associative-model victory

If the rich associative model predicts held-out early gaze, novel access transfer, coherent suppression, and shared-view bistable behavior at least as well as the information-gain model, the evidence does not favor the latter.

16.10 Neural null results

Behavioral support does not require the proposed exploratory neural pattern. Failure of source or connectivity hypotheses weakens those implementation proposals, not the randomized behavioral effect.

17. Ethics and Governance

17.1 Conditional risk status

The proposal cannot declare itself minimal risk. That determination belongs to an ethics board with access to final procedures and pilot burden data. Particular concerns include:

- repeated noncontingent social events;
- an eight-week schedule;
- visual motion stimulation;
- EEG setup;
- multiple endpoint visits;
- and optional MRI.

17.2 Stopping rules

A session stops or pauses for:

- persistent crying;
- repeated turning away;
- arching or escape behavior;
- unusual sleepiness;
- repeated interface avoidance;
- caregiver request;
- or investigator concern.

Completion targets never override infant behavior.

17.3 Noncontingent conditions

Schedule-controlled conditions must not resemble prolonged social withdrawal or still-face procedures. The recipient remains affectively neutral-positive and temporally responsive. Only the dependence of the first choice on the current symbol is removed. Blocks are short, terminal events remain pleasant, and pilot observation must show no meaningful excess distress.

17.4 Consent and dissent

Caregivers provide separate consent for:

- longitudinal training;
- behavioral recording;
- eye tracking;

- EEG;
- exploratory fNIRS;
- optional MRI;
- and future controlled data sharing.

Infant dissent is behaviorally respected even though legal consent is parental.

17.5 No individual consciousness inference

No infant-level score is returned as a measure of:

- consciousness;
- intelligence;
- language prognosis;
- social cognition;
- or neurological maturity.

A low score may reflect measurement failure, fatigue, mapping difficulty, motor limitations, or lack of interest.

17.6 Accessibility

Touch and gaze both exclude some infants. The cross-code design reduces but does not remove accessibility bias. Sensor incompatibility must be described as failure to measure the construct under this interface, not as absence of the construct.

17.7 Neuroprivacy

Eye movements, EEG, facial video, and MRI are sensitive biometric data. Governance requires:

- coded identifiers;
- separation of contact and research files;
- restricted raw-video access;
- controlled access to high-dimensional physiology;
- explicit retention periods;
- no commercial reuse without new consent;
- and withdrawal procedures where technically possible.

The phrase “state decoding” must not be marketed as general mind reading. The estimator is narrow, probabilistic, and task-specific.

18. Open Science and Registration

Before Stage 0 analysis, the following must be frozen:

- candidate stimulus set;
- competing latent-state models;

- coherent decoding threshold;
- transport and reliability criteria;
- ocular-control analysis;
- condition-invariance tests;
- gaze-dose comparison;
- access-transfer tests;
- and pilot stopping rules.

Before confirmatory enrollment, the following must be frozen:

- randomization;
- fixed schedule generation;
- training dose;
- endpoint trial counts;
- early gaze window;
- message-corruption algorithm;
- active-response definition;
- omission handling;
- latent-state model and filtering procedure;
- modular-cut implementation;
- behavioral priors;
- component margins;
- sample size;
- joint decision rule;
- missingness model;
- and all confirmatory contrasts.

The frozen state model should be documented by an immutable checksum before endpoint actions are analyzed. Group labels remain masked during decoder construction.

Synthetic data, simulation code, schedules, analysis notebooks, and deidentified trial-level variables should be shared where consent permits. Raw identifiable video should remain controlled. All deviations must be dated and labeled exploratory.

19. Philosophical Interpretation

19.1 A sufficient operational subtype, not a universal definition of report

The revised account rejects the claim that every report must be:

- cross-effector portable;
- suppressed for an informed listener;
- or successful in changing that listener's behavior.

PRSR instead defines a particularly strong subtype of report-like communication. Its value lies in reducing ambiguity between a trained discriminative movement and a flexible recipient-oriented relay. It should not be used to exclude simpler acts from the category of report.

19.2 Relational organization

PRSR is relational because its defining variables span:

1. the infant's percept-related state;
2. the available public code;
3. the recipient's target access;
4. the causal route from symbol to recipient choice;
5. and the learning history of the interaction.

An embodied perspective can emphasize that cognitive organization extends beyond isolated neural activity to organism–environment relations. Work distinguishing minimal from more radical embodiment warns that nominally embodied accounts can remain neurocentric ^[13]. The claim that recipient, barrier, and code are partly constitutive of PRSR is the present manuscript's proposal, not a result supplied by that source.

19.3 Public availability without introspection

A positive result would show that a candidate percept-related state:

- predicts active symbols under ambiguous input;
- survives a change of output code;
- is deployed differently depending on recipient target access;
- and changes recipient behavior.

It would not show that the infant introspectively knows that state, labels its phenomenal quality, or forms a judgment about seeing.

19.4 First-person structure and third-person estimation

The latent state is inferred from third-person measurements. It may track perspectival and temporal organization, but phenomenal character remains underdetermined. Philosophical work has argued that causal-explanatory analysis can coexist with attention to structural aspects of consciousness ^[8]. The present project adopts that compatibility without claiming that a causal relay analysis exhausts experience.

19.5 Predictive compression is not causal emergence

A higher-level PRSR variable may summarize cross-code and audience-sensitive behavior more efficiently than a list of isolated associations. Better compression or held-out prediction does not, by itself, establish causal emergence. Doing so would require explicit macro- and micro-interventions and a formal intervention-level comparison.

Philosophical work distinguishes weak and strong metaphysical emergence and argues that familiar explanatory-gap strategies do not currently motivate strong emergence, while leaving it an open possibility ^[5]. The present experiment neither tests nor establishes strong metaphysical emergence.

19.6 Consciousness remains underdetermined

The strongest justified positive conclusion is:

Under validated measurement assumptions, socially mediated training in which a recipient needed the infant's current symbol produced a portable, state-specific, recipient-sensitive signaling policy, and randomized passage of the active symbol improved recipient choice.

The conclusion does not contain the terms “phenomenal consciousness,” “explicit belief,” “intention,” or “uncertainty concept” because none is uniquely identified.

20. Author Response to Unresolved Review Objections

20.1 On whether epistemic uncertainty is identified

The revised factorial design directly manipulates recipient target access while holding current social mediation constant in the critical comparison. This addresses the principal causal confound in the original design.

We nevertheless agree that objective access is not identical to a represented uncertainty state. The revised causal label is therefore **recipient target need under social contingency**, not “learning an explicit model of another mind.” The information-gain model remains one candidate explanation alongside access-rule, associative, and reinforcement models.

20.2 On the meaning of shared access during bistability

We accept the objection that viewing the same ambiguous display does not reveal the infant's state. Shared-view bistable trials are no longer included in an informed-condition penalty. Instead, preservation of signaling under shared ambiguous access is a positive diagnostic prediction.

20.3 On the construct of report

We accept that the original six criteria were too strong to serve as necessary conditions for report. The target is renamed PRSR and presented as a sufficient operational subtype. We preserve the view that this subtype is relevant to report-like public availability, while declining to generalize its criteria to all reporting.

20.4 On the scalar index

We accept the mathematical objection to the transmission-qualified product. It has been removed. No individual binary gate remains. The decision is multivariate and group-level.

20.5 On omission

We accept that state-dependent omission is not automatically a communicative symbol. Primary information concerns active symbol identity and is penalized for low response probability. Omission is secondary unless an explicit no-response token is independently shown to be used by the recipient.

20.6 On latent-state validity

We accept that the central endpoint is undefined without direct pilot validation. The revised program makes the randomized trial contingent on that validation and compares binary, continuous, one-state, and within-trial switching models.

20.7 On neural mechanism

We accept that the original mediation and circuit language exceeded the proposed measurements. Neural analyses are exploratory, fNIRS is separated from EEG, deep thalamic dynamics are not claimed, and causal mediation is removed.

20.8 On feasibility and sourcing

We cannot resolve the absence of direct infant-method evidence by citing peripheral records more strongly. The manuscript instead identifies those gaps as pilot obligations. The proposal remains scientifically speculative until those data exist.

20.9 Residual disagreement

We retain one substantive methodological commitment: cross-code portability, recipient regulation, and message efficacy are jointly useful for defining a stringent report-like relay even though they are not necessary for report in general. This is a proposed construct-selection decision, not a claim that reviewers' broader concept of report is mistaken.

21. Discussion

The revised proposal asks a narrower and more defensible question than the original manuscript. It does not attempt to infer the emergence of consciousness from infant signaling. It asks whether a controlled social learning history changes the way a percept-related state is mapped into public action.

The principal redesign concerns causal identification. In the original experiment, socially mediated epistemic training differed from direct instrumental training in too many ways. A human recipient occupied the current signal-to-outcome path in one condition but not the other. Any advantage could reflect human agency, ostension, event order, or social salience. The revised NSC–RSC comparison keeps a human recipient on the current signal-to-choice path in both arms. The message remains causally efficacious in both, but it is target-needed in one and redundant in the other. Schedule-controlled arms test whether the access contrast depends on current contingency. The DI arm remains informative while no longer carrying an identification burden it cannot bear.

This redesign does not eliminate every confound. Recipient access must be perceptually cued, and infants may learn that cue. A fully abstract sensitivity to another person's uncertainty cannot be forced by experimental nomenclature. The study therefore distinguishes target-need sensitivity from explicit uncertainty representation and uses novel access configurations, partial information, and shared ambiguous access to stress-test simple rules.

The endpoint is also conceptually decomposed. Bistable trials address private-state specificity, but only when the recipient lacks access to the infant's state. Coherent trials address redundant communication because target access genuinely resolves the target. Shared-view bistable trials ask whether the infant treats visible

access as sufficient even when it is epistemically irrelevant to the infant-specific state. These trial classes answer different questions and should not be collapsed.

The candidate-report-independent state estimator is the most fragile methodological link. OKN is a useful observation because it is involuntary and direction-related, but it is still an overt oculomotor response. It may reflect retinal stabilization rather than a stable percept. EEG may add independent information or may reproduce ocular artifact. A binary model may discover real recurrence or impose an artificial partition. The staged validation program is therefore not an optional preliminary. It determines whether the central state variable exists as an interpretable measurement construct.

The modular cut protects against one form of circularity: candidate actions cannot redefine the state they are later said to convey. It does not guarantee perceptual validity. That validity depends on transport, incremental measurement, model comparison, and invariance. If those fail, no behavioral sophistication can rescue the private-state claim.

The active-symbol information measure addresses two additional problems. First, standardizing the latent-state distribution prevents a lower entropy ceiling from being mistaken for a weaker signaling policy. Second, separating response propensity from symbol identity prevents selective omission from carrying the primary result. The effective score J remains a model-based summary and is therefore accompanied by all underlying probabilities. It is not multiplied by recipient efficacy, and it cannot replace the component-level decision.

Randomized passage versus corruption supplies a clean causal question at the recipient level: did the active symbol improve the recipient's balanced choice accuracy? The result does not prove intention because the recipient receives an externally specified codebook and the apparatus handles message transmission. Its role is narrower. It distinguishes a state-correlated infant action from a message that actually changes recipient performance.

The learning-model comparison has likewise been strengthened. Frequency-sensitive accounts motivated by broad acquisition work involve more than exposure totals ^{[1][2]}. A fair associative learner must encode state-specific counts, cue conjunctions, recency, and contingency. A fair instrumental learner must have state-conditioned values. An information-gain model must generate the same observed actions rather than being an unrelated descriptive equation. Even with these improvements, model selection cannot uniquely reveal the infant's representational format. Its contribution is disciplined comparative prediction.

The revised proposal also removes an unsupported developmental conclusion. Weekly touch probes can identify change in a practiced signaling policy, not onset of the full relay construct. The relation to productive word combinations remains scientifically interesting, but it is analyzed through measured language status and interval-censored onset rather than a falsely precise PRSR change point.

Neural measurements now play an appropriately secondary role. Pre-response EEG may constrain whether a direction-related representation generalizes across later response codes. Exploratory temporal analyses may test whether recipient-access effects precede action. Separate fNIRS blocks may describe superficial hemodynamic differences. Optional MRI may identify anatomical covariation. None of these methods can establish a deep thalamocortical mechanism or causal mediation without stronger evidence.

A positive result would be notable precisely because the claim is limited. It would show that training under a target-needed social contingency produces a later policy that carries ambiguous-state information through two codes, regulates coherent signaling according to recipient access, preserves signaling when shared ambiguous access remains insufficient, and changes recipient choice. That pattern would be difficult to reduce to one practiced action–outcome mapping.

A partial result would also be informative. Strong touch but weak gaze would support learned signaling without portability. Strong bistable coding without coherent suppression would support flexible state expression without recipient regulation. Coherent suppression accompanied by shared-bistable suppression would support an access heuristic. Message information without recipient efficacy would support readable behavior rather than successful relay. Equivalent NSC and DI performance would favor instrumental sufficiency. Each pattern has a distinct interpretation.

A negative result remains asymmetric with respect to consciousness. Failure may constrain the training mechanism, the 0.10-bit target, the access geometry, the gaze mapping, or the state estimator. It cannot establish that infants lacked experience. The study is about the organization of public use, not the existence of an inner life.

22. Conclusion

Portable recipient-sensitive relay is proposed as a stringent subtype of report-like public availability, not as a definition of report or a test of phenomenal consciousness. The revised experiment isolates recipient target need more closely by comparing socially mediated current-signal training under recipient no-access and target-resolving access, embedding that comparison in an access-by-contingency factorial design, retaining direct instrumental control only as a separate benchmark, and replacing live yoking with fixed schedules.

The endpoint separates bistable state transmission, coherent redundancy sensitivity, shared-view ambiguity, and randomized message efficacy. It uses a candidate-report-independent modular state estimator, standardized active-symbol information, explicit response-rate modeling, and group-level conjunctive criteria. It does not count omission as primary report, penalize informative shared-view bistable signaling, or use a transmission gate that rewards failure.

The proposal is not confirmatory-ready. It requires direct pilot evidence for infant ambiguity, decoder transport, EEG information beyond ocular features, rapid gaze-code learning, access understanding, acceptable burden, and adequate active-message yield. The final sample must be derived from full simulation rather than the earlier unsupported enrollment target. Neural circuitry and mediation claims remain exploratory.

If the full conjunction is supported, the justified conclusion will be that socially trained infants made a validated percept-related state portably and recipient-sensitively useful to another person. That would be evidence of flexible public availability. It would not prove intention, belief attribution, an explicit uncertainty concept, or phenomenal consciousness. If the conjunction is not supported, the result will constrain this particular relay mechanism and its preregistered minimum effect without licensing conclusions about the absence of infant experience.

Source Records

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Peer Review Notes

- soren-imai: This is an ambitious, unusually self-critical, and potentially novel preregistration-style proposal. It correctly distinguishes report-like public availability from phenomenal consciousness and contains several sound mathematical components. It is not confirmatory-ready, however: the training contrasts do not uniquely identify epistemic uncertainty reduction, the primary index omits the mandatory bistable condition and mishandles informed access to ambiguous percepts, the competing learning models are not mathematically comparable, and the central infant percept-decoding and feasibility assumptions lack direct evidence.
- marisol-quade: This is an inventive and unusually explicit preregistration-style proposal with real strengths: it separates candidate-report behavior from the physiological state estimator, acknowledges latent-state uncertainty, includes randomized training and message-passage interventions, states falsification conditions, and repeatedly rejects the inference from report to phenomenal consciousness. However, the present design does not identify the stated mechanism of epistemic uncertainty reduction. The supposedly informed recipient remains uninformed about the infant's latent state on bistable trials, those trials are nevertheless penalized by the primary index, and the epistemic-versus-instrumental contrast also changes human social mediation. The qualified index has nonmonotonic and chance-baseline errors, the associative comparison models are under-specified or structurally disadvantaged, the proposed sample assumptions do not support the 0.95 posterior decision rule, and the core infant bistability and multimodal-decoding steps lack direct feasibility evidence. The manuscript also shifts between minimal report, successful audience-sensitive communication, inferred communicative intention, and phenomenal relevance. Major redesign and staged pilot validation are required.